

OPERATIONAL STANDARD OPERATING PROCEDURE • EMERGENCY RESPONSE DIRECTIVE

11.5-DAY DISTRIBUTED EMERGENCY SPRINT PROTOCOL

Standard Operating Procedure governing autonomous execution, asynchronous swarm collaboration, and strict verification gates leading to the Smart Horizon 48-Hour International Hackathon Grand Finale under institutional workspace denial.

USABLE PREP WINDOW

11.5 DAYS

Aug 21 PM → Sept 1 (23:59 IST)

PHYSICAL LAB ACCESS

DENIED (0%)

100% Remote / Virtual Protocol

TRAVEL EXCLUSION

SEPT 2

Zero-Code Transit to NHCE

GRAND FINALE

SEPT 3-5

48h Live Offline Event

01 OPERATIONAL CONTEXT & MANAGEMENT CONSTRAINTS



CRITICAL OPERATIONAL BOTTLENECK: INSTITUTIONAL COLLOCATION DENIAL

Institutional management has formally refused on-premise physical workspace and laboratory access for the SUTRA team. Consequently, physical team collocation is prohibited during the remaining 11.5 prep days. All 6 subsystems must coordinate via a zero-friction remote async protocol with mandatory daily merge war-rooms to prevent cross-module divergence.

02 VIRTUAL SWARM OPERATING CADENCE

01. MORNING ALIGNMENT

SYNC

09:00 - 09:20 IST

- 3-Minute Hard Limit per member (Yesterday Delivered / Today Goal / Hard Blockers).
- Mandatory Rule 0 Pull: Pull latest integration changes (git fetch origin dev && git merge origin/dev).
- Pairing Assignment: Assign partner for cross-module integration tasks.

02. NIGHTLY INTEGRATION WAR-ROOM

21:30 - 22:15 IST

- Buffer Integration: Review and merge feature branches strictly into dev.
- Live Verification Gate: Run pytest test suites and npm run build.
- Zero Mock Benchmark Rule: Prohibit synthetic numbers in DOCS.md updates.

PROTOCOL RULE	OPERATING REQUIREMENT	ENFORCEMENT MECHANISM
30-Min Blocker Escalation	No developer may remain blocked on dependencies/bugs for >30 minutes without requesting a pair session.	Immediate Discord Live Share / screen pairing.
Branch Isolation	Work strictly within feature/subsystem-*. Direct commits to main or dev are blocked.	Pull Request review during 21:30 War-Room.
24/7 Virtual Discord Lab	Persistent voice room active during sprint hours. Muted by default; un-mute for instant zero-latency pairing.	Continuous presence during deep focus blocks.

03

11.5-DAY TACTICAL SPRINT BREAKDOWN

PHASE 1 • 4.0 DAYS SUBSYSTEM HARDENING Aug 21 PM – Aug 24 - Resolve NumPy ABI / TensorRT import crashes in Subsystem C. Sat, Aug 22: Attend official NHCE Pre-Event Online Briefing. - PX4 Offboard setpoint generator & SwarmRAFT unit tests. - WebGPU GCS frontend WebSocket subscriber connection. - NDMA rescue profile definitions in docs/conops/.	PHASE 2 • 4.0 DAYS SWARM INTEGRATION Aug 25 – Aug 28 - Close Loop 1: GNC flight ↔ Camera ↔ Mesh ↔ Perception. - Close Loop 2: GCS 3D GIS visualization + 1-Click RTL trigger. 5-Drone Gazebo Sim 8 world with ORCA 3D collision avoidance. - SwarmRAFT leader failover test under simulated node crash. - Record high-res backup simulation demo video reels.	PHASE 3 • 3.5 DAYS AUDITS, REHEARSAL & FREEZE Aug 29 – Sept 1 (18:00) - Execute live Gate Audits G1–G6 verification. - Update DOCS .md strictly with real measured benchmarks. - Conduct 2 timed Mock Presentations & Jury Grill Rehearsals. Sept 1, 18:00 IST: Hard Code Freeze & tag v1.0.0. - Flash 3 redundant offline USB drives (weights, npm, Docker).
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SUBSYSTEM OWNERSHIP & ACCOUNTABILITY MATRIX

LEAD MEMBER	ASSIGNED SUBSYSTEM	MANDATORY DELIVERABLES (BY SEPT 1 FREEZE)	VERIFICATION SUITE
Nikhil	SUBSYSTEMS A + B GNC, Comms & Sim	PX4 50Hz offboard setpoints, ORCA 3D collision avoidance, SwarmRAFT consensus, Gazebo Sim 8 digital twin.	pytest sutra_gnc/ pytest sutra_comms/
Vedanth	SUBSYSTEM C Edge Perception	YOLOv8–Nano TensorRT edge detector (<5ms), Tri-Modal fusion, DEM terrain raycast WGS84 target geolocator.	pytest sutra_perception/
Siva	SUBSYSTEM D 3D GIS GCS	WebGPU 60 FPS HUD, 3D Mapbox satellite drone tracking, WebSocket telemetry listener, Emergency 1-Click RTL.	npm run build (in sutra_gcs/)
Harika	SUBSYSTEM E Docs & Audits	Automated test suites, G1–G6 benchmark audit sync in DOCS .md, Master Pitch Deck visual formatting, PDF packages.	pytest full run & doc audit
Rohith	SUBSYSTEM F Tactical Ops	NDMA search patterns (Kedarnath flood / Wayanad landslide), Pre-flight SOP, operational rescue storytelling narrative.	CONOPS audit in docs/conops/

05

TRAVEL & ON-SITE OFFLINE BATTLECARD (SEPT 2 – SEPT 5)

SEPT 2: TRAVEL DAY PROTOCOL (ZERO-CODE) - Travel to Bengaluru (NHCE Campus) — mandatory physical rest. - Inventory Check: High-wattage power strips, multi-plugs, mobile Wi-Fi hotspots. - Distribute 3x offline USB flash drives containing entire cached codebase and simulation reels.	SEPT 3–5: 48-HOUR LIVE OFFLINE EXECUTION - Zero dependency on venue Wi-Fi: Run all ROS 2 nodes and GCS on local loopback/offline network. - Deliver live 5-drone Gazebo Sim 8 swarm demonstration with WebGPU HUD telemetry. - Execute NDMA rescue mission narrative during jury evaluation rounds.
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COMPLIANCE & INTEGRITY GOVERNANCE All operations must adhere strictly to the Master Autonomous Agent Protocol (AGENTS.md). Synthetic, mock, or projected benchmark figures are strictly prohibited across all documentation and audits. By executive order of the Tech Architect, this SOP takes immediate legal and operational effect.	✓ OFFICIALLY AUTHORIZED PROJECT SUTRA • 2026
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